Plasma Science & Fusion Center

Making Real Machines Work: Modelling with States, Actions and Observations

Allen M. Wang

Starting a Non-Fusion Job Soon, Might Lack Some Hinges

Session Format

- Mostly centered around slides
- Will jump into notebook(s) periodically
- Targeting lots of audience engagement

Overview

- Motivation for a shift in perspective
- Intro to modelling with states, actions, and observations
  - Crucial to ideas like state-space models, reinforcement learning (RL)
  - Key outcome: even w/ perfect physics fidelity, there are challenges
- Intro to learning dynamics from data
  - The classical perspective
  - The F=ma + AI perspective
- Modelling without physics state
  - Learned state representations
  - Action-conditioned world models
- Some comments on AI agents
  - The code in this tutorial was almost entirely vibe-coded
  - It’s important to pay attention to what is happening

Motivating a Shift in Perspective

A Crucial Shift in Perspective

Physics

"What is true and invariant in general?"

Physics is about constructing conceptual thoughts that are invariant

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\cdot\frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{q_\alpha}{m_\alpha}(\mathbf{E}+\mathbf{v}\times\mathbf{B})\cdot\frac{\partial f_\alpha}{\partial \mathbf{v}} = 0$$

$$\nabla\cdot\mathbf{B} = 0$$

$$\nabla\cdot\mathbf{E} = \frac{1}{\epsilon_0}\sum_\alpha \overline{n}_\alpha q_\alpha \int f_\alpha\, d\mathbf{v} + \frac{\rho_{q\,\mathrm{ext}}}{\epsilon_0}$$

$$\nabla\times\mathbf{B} = \mu_0\sum_\alpha \overline{n}_\alpha q_\alpha \int \mathbf{v}\, f_\alpha\, d\mathbf{v} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} + \mu_0\mathbf{J}_{\mathrm{ext}}$$

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$$

α stands for ions and electrons.

$$\Delta^*\psi \equiv r\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\psi}{\partial r}\right) + \frac{\partial^2\psi}{\partial z^2}, \quad F(\psi) \equiv rB_\phi$$

$$J_\phi^{plasma} = r\frac{dp(\psi)}{d\psi} + \frac{F(\psi)}{\mu_0 r}\frac{dF(\psi)}{d\psi}$$

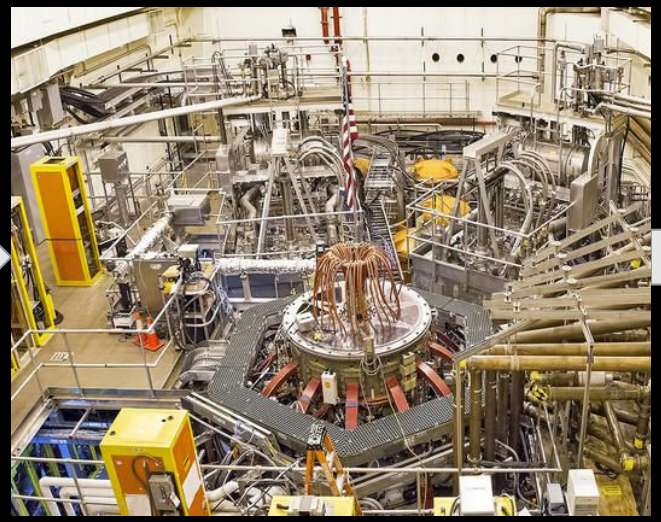
$$\Delta^*\psi = -\mu_0 r(J_\phi^{plasma} + J_\phi^{vacuum})$$

Controls + Ops

"If I poke the thing with volts at time 't', what volts do I see at 't+dt'?"

We must make the best choice in the moment, given what we can see.

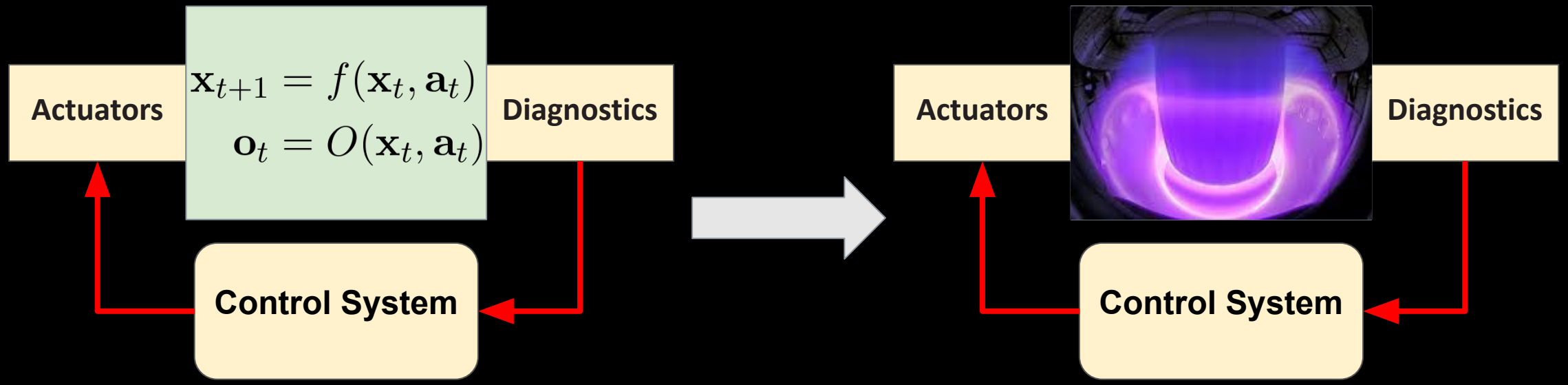
Volts



Volts

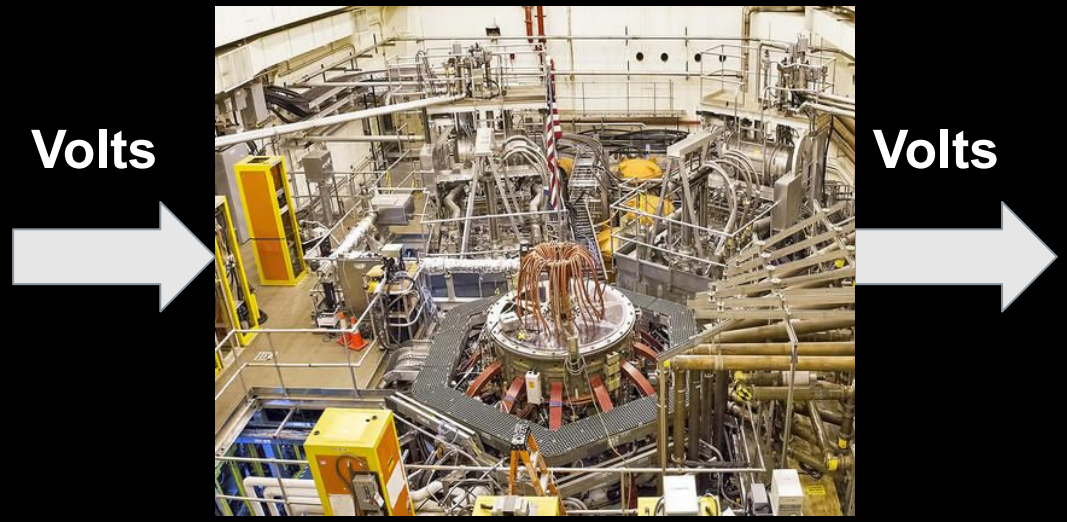
The Importance of Modelling for Control

- Tokamak trials are expensive + risky (at SPARC scale)
  - We need to succeed with as few trials as possible



A Crucial Shift in Perspective

- Physics-based modelling is one tool to solve the problem
- Other tools:
  - Engineering specs/data
  - Past diagnostic data
  - Design of new experiments
  - Modifying the machine
- Challenging problems demand that all useful tools are used



Stand on the Shoulders of Giants

- The aircraft control community took a data-driven approach to aerodynamic modelling many decades ago
- In 2022, computational fluid dynamics (CFD) was **still** not good enough for aircraft stability and control modelling
  - I expect physics simulation to play a big role in SPARC ops
  - If you believe plasma physics modelling is much harder than CFD, you better believe we need to invest a lot more into the other tools in ops

of Reynolds-averaged Navier Stokes (RANS) based Computational Fluid Dynamics (CFD) to capture with sufficient fidelity vehicle forces and moments in flows dominated by these nonlinear aerodynamics. As a result, budget and schedule penalties, as well as possible envelope and performance penalties, are incurred. While the use of CFD to predict vehicle performance has matured significantly over the years, its role in Stability and Control (S&C) analysis has remained small. As pointed out in Ref. 1, there are a few reasons for this. Chief among them is that vehicle performance is generally confined to relatively benign regions of the envelope, while S&C analysis needs to consider off-design conditions, where complex aerodynamics that challenge RANS models are prevalent. This, combined with a lack of understanding of the limitations of the RANS approach by its early progenitors, led to a distrust of CFD by the S&C community.

An attempt at community consensus on the 2020 state of the art of S&C prediction was made in Ref. 4. In that

Chyczewski, Thomas S., et al. "Summary of the first AIAA stability and control prediction workshop." *AIAA SciTech 2022 Forum*. 2022.

Aircraft System Identification

$$m\dot{V} + \omega \times mV = F_{Aero} + F_{Thrust} + F_{Gravity} \qquad \left[\frac{d}{dt}(mV) = \sum F\right]$$

$$I\dot{\omega} + \omega \times I\omega = M_{Aero} + M_{Thrust} \qquad \left[\frac{d}{dt}(I\omega) = \sum M\right]$$

***Typical aircraft system identification problem*:**

From measurements of the inputs and outputs, determine mathematical model forms for F_{Aero} and M_{Aero}, then estimate the unknown parameters in those models

System Identification is used to model what is not known, based on measured data

Morelli Eugene, Aircraft System Identification, NASA Langley Research Center 2011

S*** Control Engineers Say

Physics and System Identification

Scientific theories are not discoveries of the laws of nature but rather inventions of the human mind.

-- Dr. Athanasios Papoulis

All models are wrong; some are useful.

-- Dr. George Box

10

Morelli Eugene, Aircraft System Identification, NASA Langley Research Center 2011

Tokamak Magnetic Control

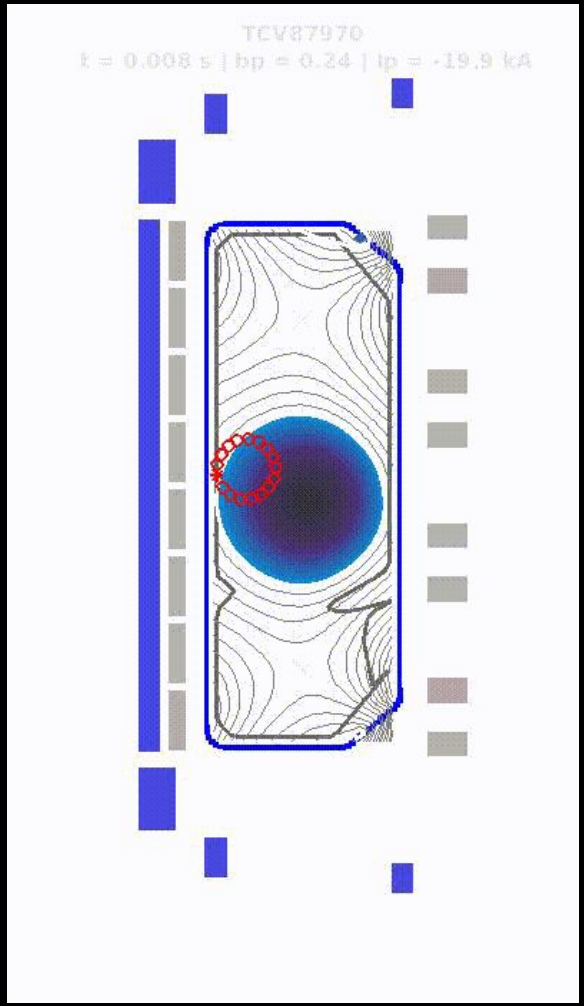
- Let us forget about physics for a moment

Power Supplies

Magnetic Probes

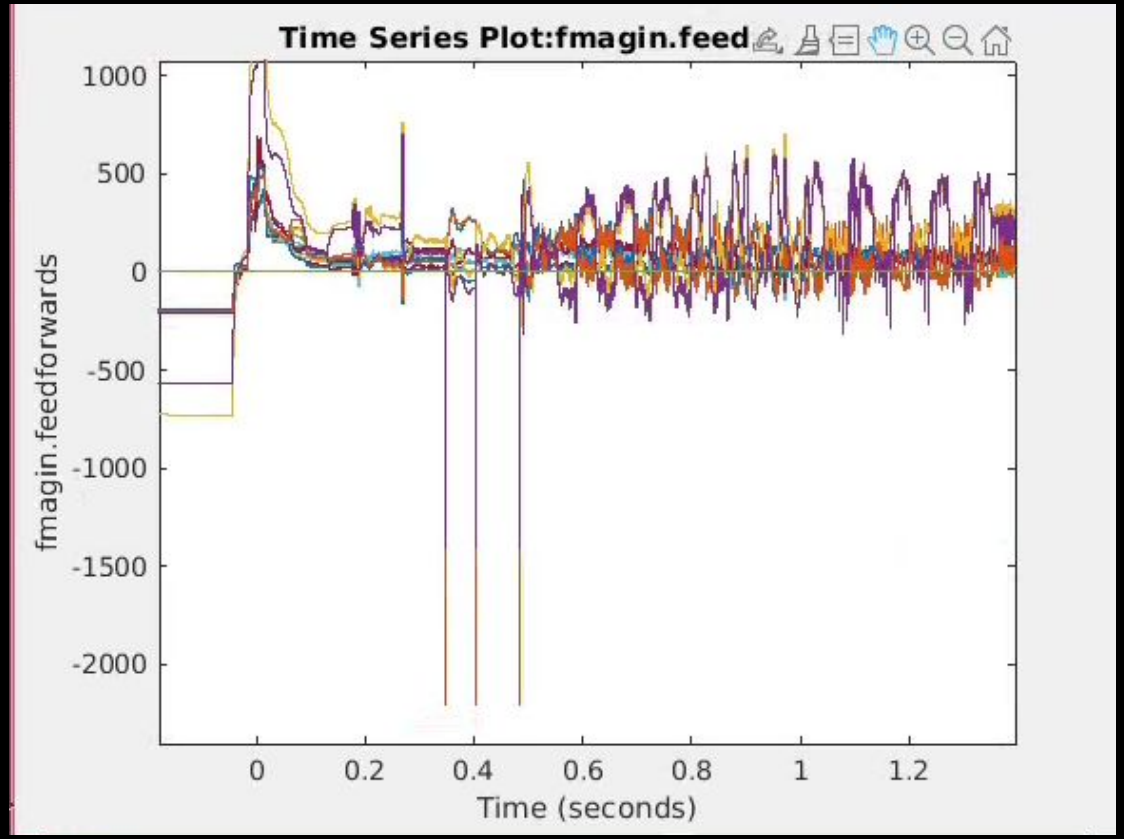
Plasma Control System

What Caused This Radial Instability?



Power Supply Voltage Spikes

Caused by 100 microsecond offset in timebase between pre-programmed and RT



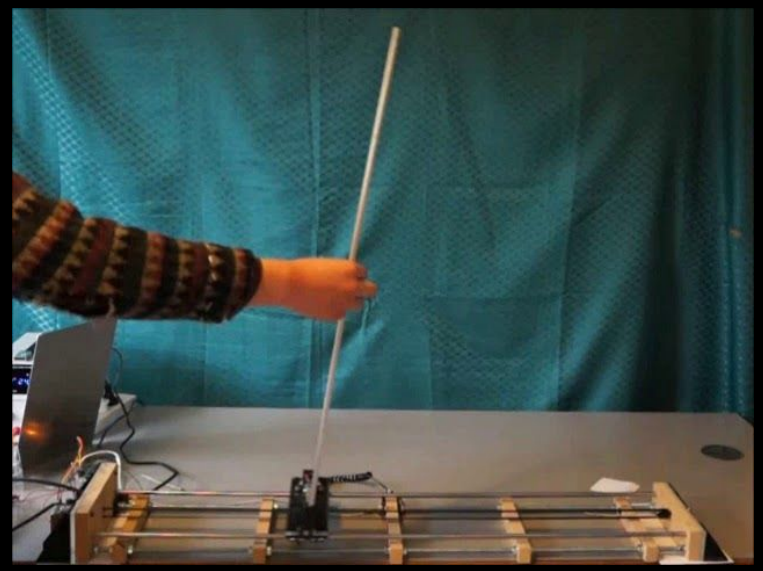
Introduction to States, Actions, Observations

Toy Example: The Cartpole

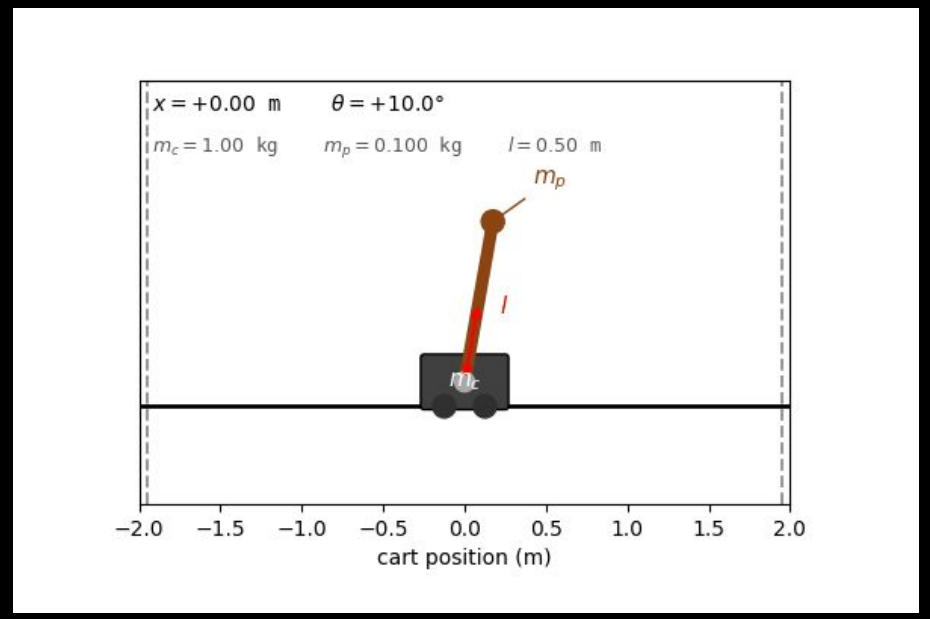
The Cartpole

- Canonical controls + RL problem
  - By applying force to the “cart”, balance the “pole”

An image of a real cartpole system

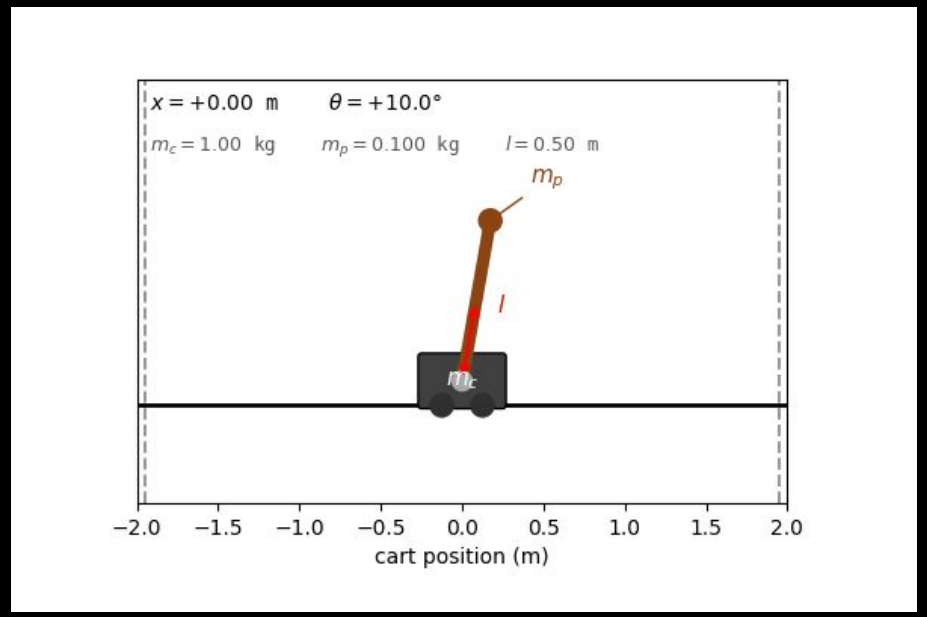


Our modelled/simulated cartpole



Cartpole Model

- The model is just Newtonian mechanics
  - At this point, everything should be familiar
- For simulation purposes, we will re-write this into a **state-space representation**



$$\ddot{\theta} = \frac{Mg\sin\theta - m_p l\,\dot{\theta}^2 \sin\theta\cos\theta}{l\left(M - m_p\cos^2\theta\right)}$$

$$\ddot{x} = \frac{m_p l\left(\dot{\theta}^2\sin\theta - \ddot{\theta}\cos\theta\right)}{M}$$

💡 Pause and Ponder

How many states are there? What are they?

$$\ddot{\theta} = \frac{Mg\sin\theta - m_p l\,\dot{\theta}^2 \sin\theta\cos\theta}{l\left(M - m_p\cos^2\theta\right)}$$

$$\ddot{x} = \frac{m_p l\left(\dot{\theta}^2\sin\theta - \ddot{\theta}\cos\theta\right)}{M}$$

"state-space form"

$$\frac{d}{dt}\mathbf{x} = f(\mathbf{x}, \mathbf{p})$$

Model parameters (mass, 'l', g)

$$\mathbf{x} \equiv \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

$$\frac{d}{dt}\mathbf{x} = \begin{bmatrix} \dot{x} \\ \dfrac{m_p l\left(\dot{\theta}^2\sin\theta - \ddot{\theta}\cos\theta\right)}{M} \\ \dot{\theta} \\ \dfrac{Mg\sin\theta - m_p l\,\dot{\theta}^2\sin\theta\cos\theta}{l\left(M - m_p\cos^2\theta\right)} \end{bmatrix}$$

Numerical Integration

- Many methods exist to time-step differential equations
  - Explicit Runge-Kutta, Implicit Runge-Kutta, Symplectic methods, etc.
- Let's think about things in terms of discrete-time going forward
  - Also helpful for handling discrete variables (e.g. is the cartpole hitting a boundary?)

💡 **Pause and Ponder**

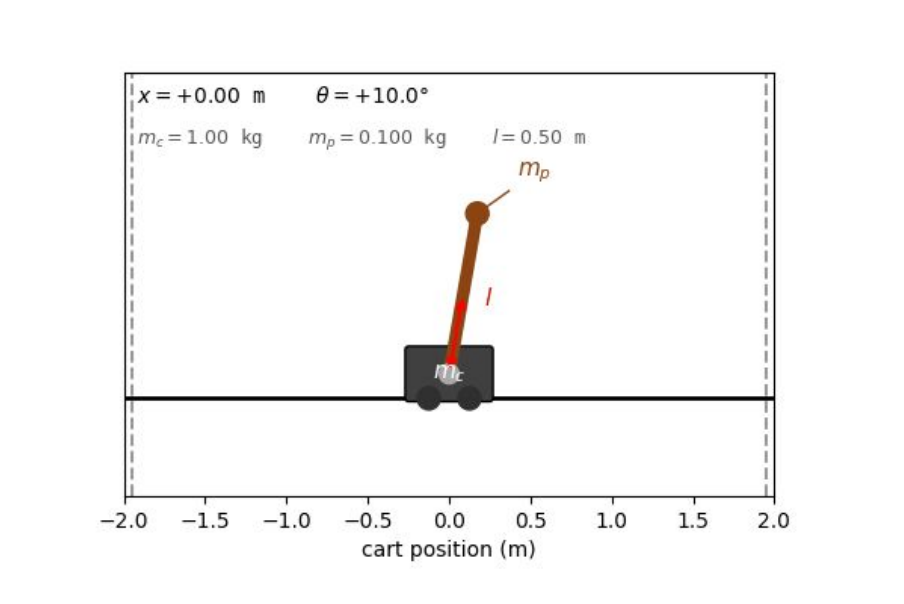
What else is needed to run a simulation?

$$\frac{d}{dt}\mathbf{x} = f(\mathbf{x}, \mathbf{p}) \quad \Rightarrow \quad \mathbf{x}_{t+1} = f_{discrete}(\mathbf{x}_t, \mathbf{p})$$

Non-Uniqueness of State-Space Representations

- There are multiple equivalent state-representations
  - E.g. we can use the position of the pole instead of “theta”
- State representations can be a design choice

$$\mathbf{x} \equiv \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} \quad \textbf{OR} \quad \begin{bmatrix} x \\ \dot{x} \\ x_p \\ \dot{x}_p \end{bmatrix}$$



Adding Actions

- Okay, great, but how do we control it?
- Let's add "Force" actions to this model

State $\mathbf{x} \equiv \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$

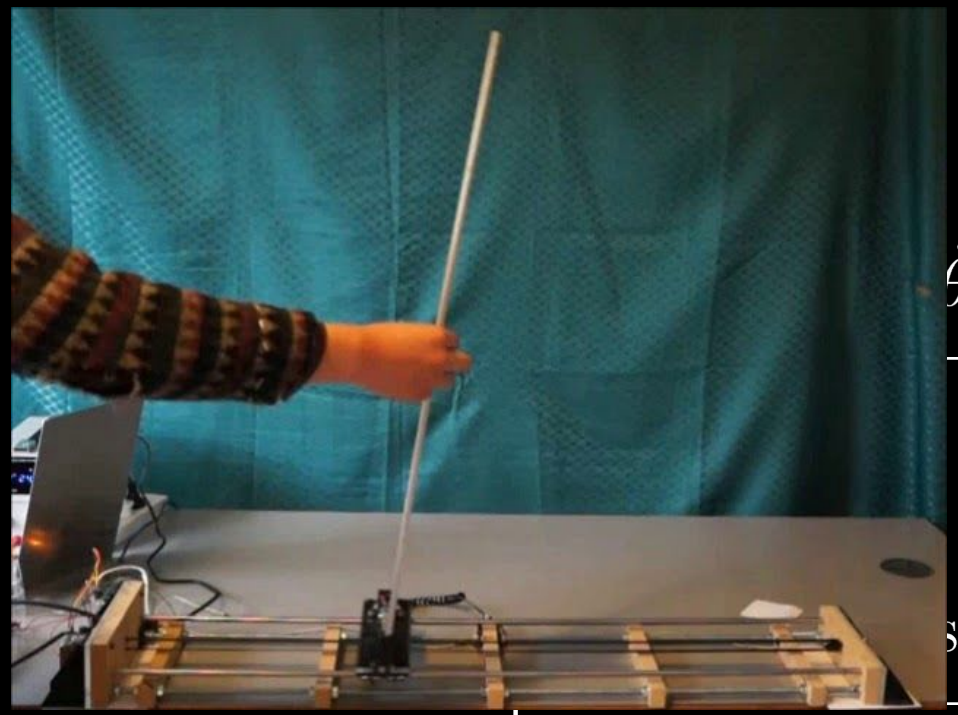
$$\frac{d}{dt}\begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dfrac{\boxed{F} + m_p l\left(\dot{\theta}^2 \sin\theta - \ddot{\theta}\cos\theta\right)}{M} \\ \dot{\theta} \\ \dfrac{M g \sin\theta - \cos\theta\left(\boxed{F} + m_p l\,\dot{\theta}^2 \sin\theta\right)}{l\left(M - m_p \cos^2\theta\right)} \end{bmatrix}$$

Action $\mathbf{a} = [F]$

Observations

- What are observables?
  - It depends on how much money you spent
- How might we observe the system state here?

State $\mathbf{x} \equiv \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$

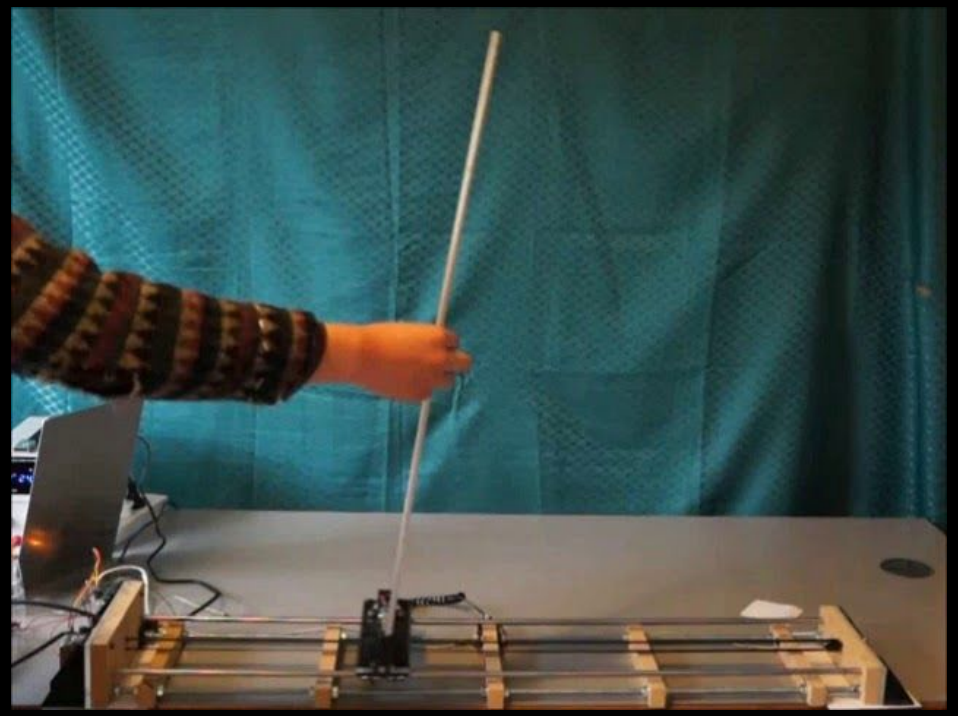


$$\begin{bmatrix} \dot{x} \\ \dot{\theta}^2 \sin\theta - \ddot{\theta}\cos\theta \Big) \\ \hline M \\ \dot{\theta} \\ s\,\theta \Big(F + m_p l\,\dot{\theta}^2 \sin\theta \Big) \\ \hline l\,(M - m_p \cos^2\theta) \end{bmatrix}$$

Action $\mathbf{a} = [F]$

Observations

- Some good answers:
  - Motor encoders estimate theta (theta_dot is noisier, why?)
  - Laser range-finders estimate x
  - Camera estimates everything (I'm serious)



Modelling Observations

- An “observation function” maps state and actions to observations

$$\mathbf{o} \sim O(\mathbf{x}, \mathbf{a}; \mathbf{p})$$

- For this exercise, let’s assume we have sensors that can measure position and angle with noise.

$$\mathbf{x} \equiv [x, \dot{x}, \theta, \dot{\theta}] \qquad \mathbf{o} \equiv [x, \theta]$$

State-Space Models and POMDPs

- Terminology often a bit of a mess
- A general nonlinear state-space model:

$$\mathbf{x}_{t+1} \sim f(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

(RL): dynamics/transition function

$$\mathbf{o}_t \sim O(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

(RL): observation function

- What is being expressed?
  - There is a set of variables that governs time-evolution (states)
  - There is a set of variables we can do something about (actions)
  - There is a set of variables we can actually see (observations)
- Another fair notation

$$\mathbf{x}_{t+1}, \mathbf{o}_t \sim f(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

State-Space Models and POMDPs

- Reinforcement learning (RL) is more or less about solving Partially Observable Markov Decision Processes (POMDPs)
- POMDP = state space model + reward function + policy

Reward function: won starcraft/plasma disrupted/theorem proved

$$\max_{\pi} \quad \mathbb{E}\left[\sum_{t=0}^{\infty} r(\mathbf{x}_t, \mathbf{a}_t)\right]$$

$$\mathbf{x}_{t+1} \sim f(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

$$\mathbf{o}_t \sim O(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

$$\mathbf{a}_t = \pi(\mathbf{o}_t)$$

Policy: maps observations to actions

POMDPs

PomDP the PhD Rapper - MIT

30K views • 5 years ago

$$\max_{\pi} \quad \mathbb{E}\left[\sum_{t=0}^{\infty} r(\mathbf{x}_t, \mathbf{a}_t)\right]$$

$$\mathbf{x}_{t+1} \sim f(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

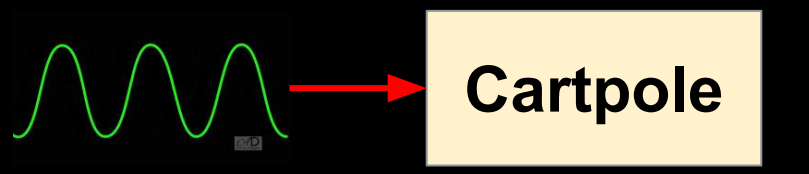
$$\mathbf{o}_t \sim O(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

$$\mathbf{a}_t = \pi(\mathbf{o}_t)$$

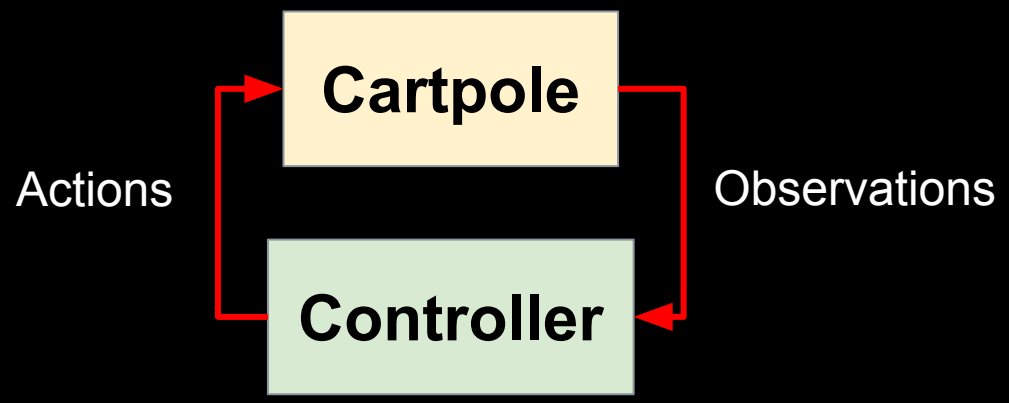
✏️ Exercise — Feedforward Control

Feed-forwards and Feedback

Feed-forwards



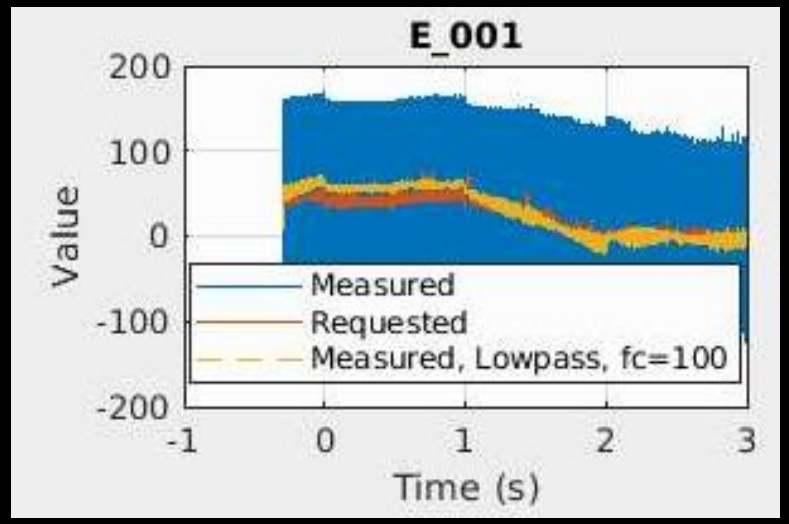
Feed-back



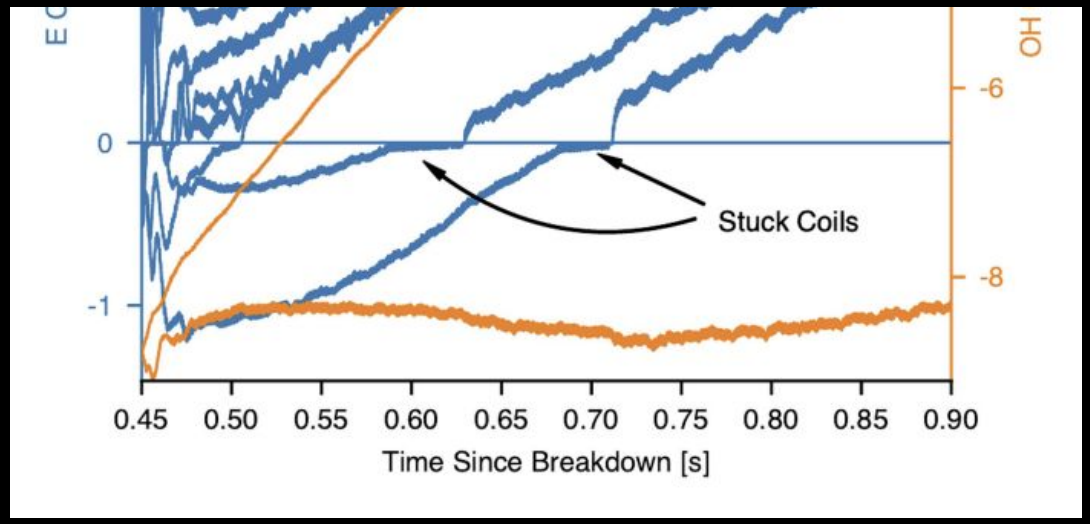
Pause and Ponder

- **Is it reasonable to say we can command Force?**
  - Not entirely
  - How reasonable depends on your hardware
  - Actuator modelling is often **very** important

Example requested vs. measured voltage on TCV power supplies

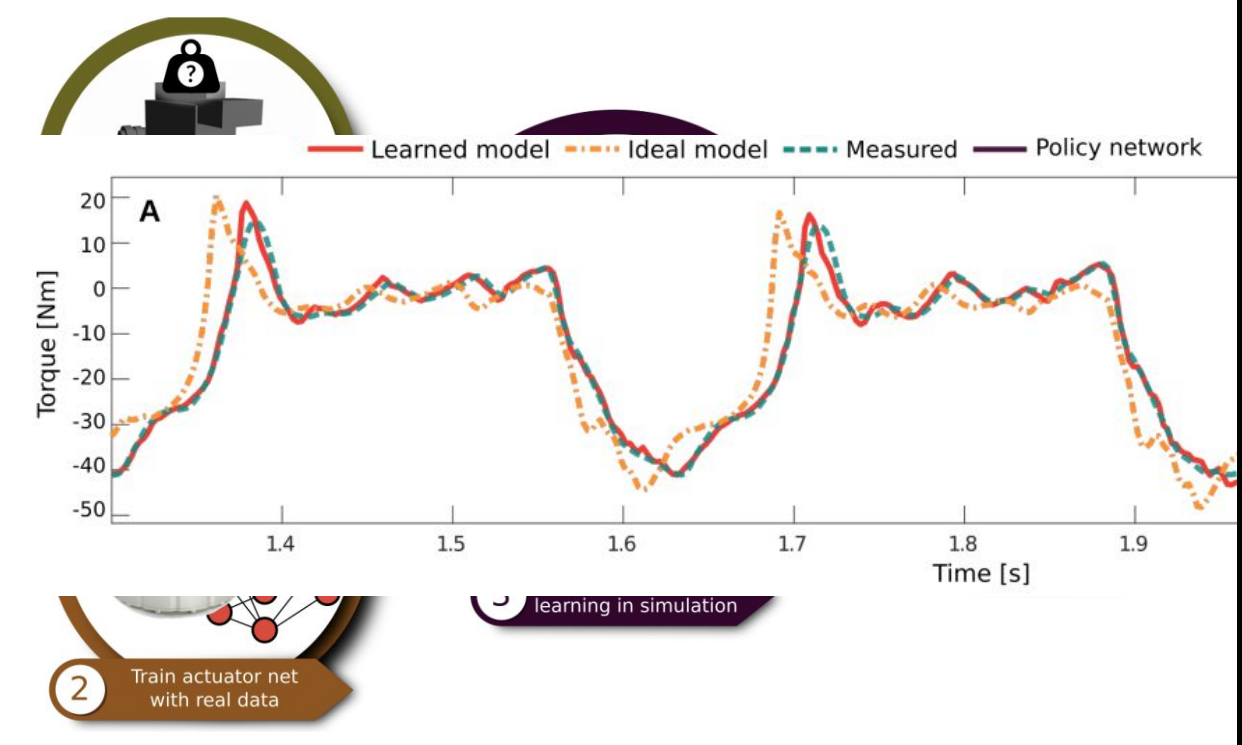


DeGrave et al., 2022 Nature



Hardware Modelling is Sometimes Critical

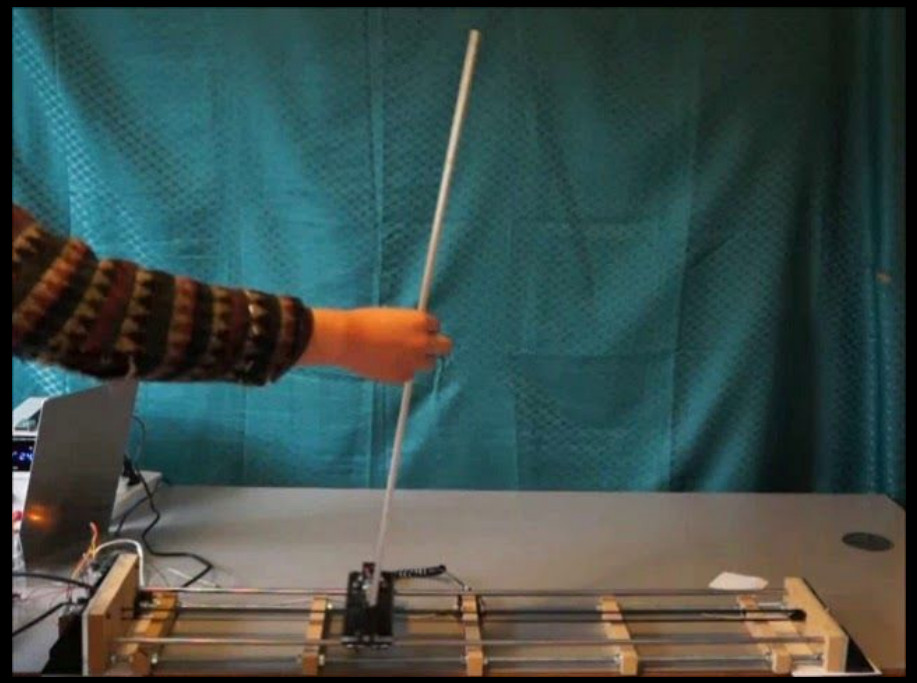
- After AlphaGo, lots of people started working on RL for controlling robots
- Folklore from D. Bellicoso: they (a team at ETH Zurich) succeeded first because they realized you need to learn a model of the motors



Learning agile and dynamic motor skills for legged robots 2064 2019
J Hwangbo, J Lee, A Dosovitskiy, D Bellicoso, V Tsounis, V Koltun, ...
Science robotics 4 (26), eaau5872

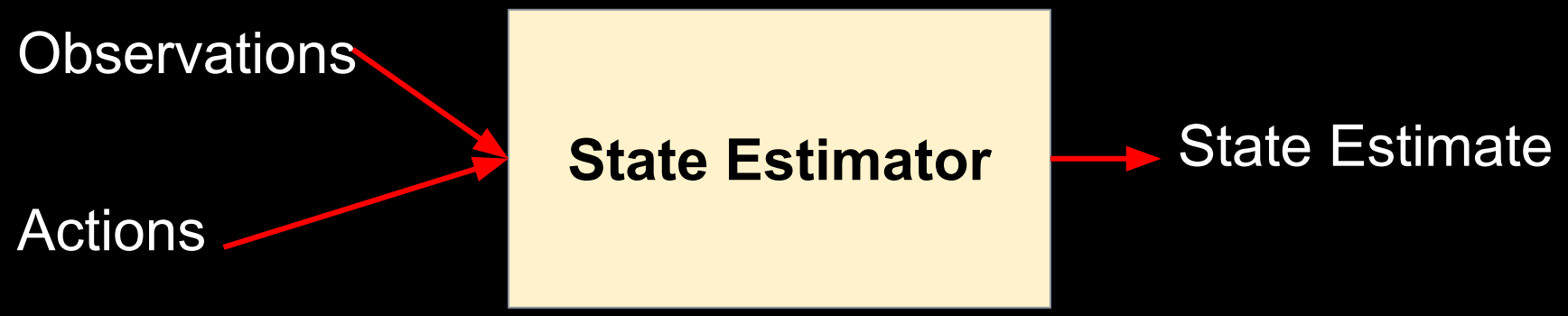
💡 Pause and Ponder: Real2Sim

- Consider the real cartpole below. We want to use our simulator to predict what it will do.
  - How might we go about doing this?
  - What challenges might we encounter?



State Estimation

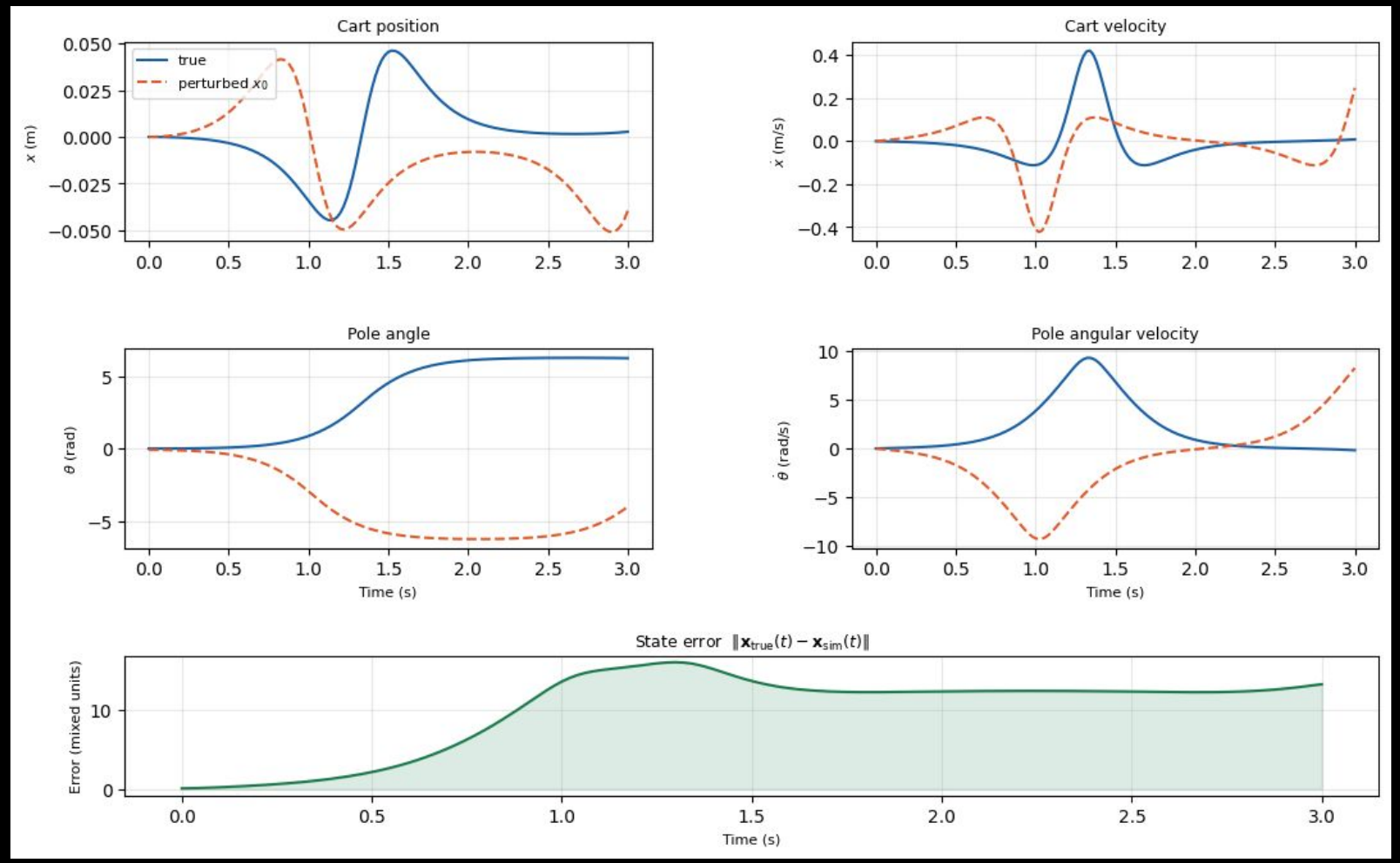
- We never know what the real system state is
- Using observations and actions to determine state is the problem of **state estimation**



✏️ Exercise — Sensitivity to Initial Conditions

```
# — TODO: CHANGE ME ————————
theta_error = jnp.deg2rad(
    -5.0
)  # TODO: change me to non-zero
theta_dot_error = jnp.deg2rad(
    -1.0
)  # TODO: change me to non-zero

x0_measured = x0_true + jnp.arra
```

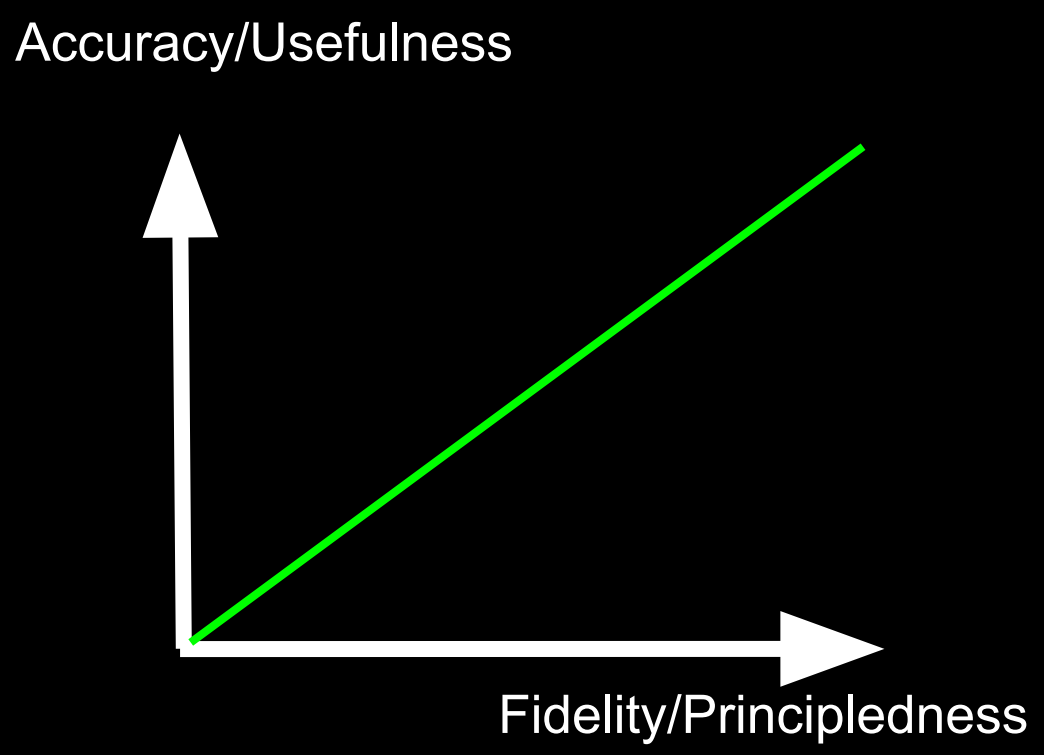


🤔Simulation Implications

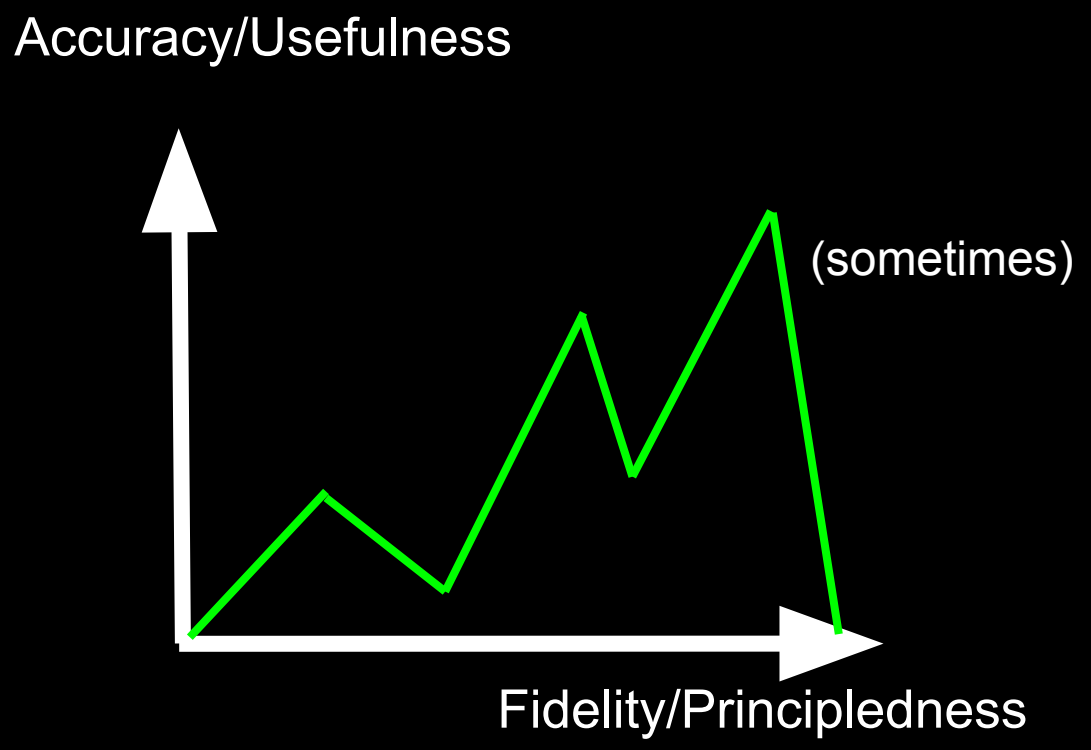
- We had perfect sim fidelity… but our predictions failed!
- All it took was a single minor imperfection: a small amount of measurement error
- Common sim challenge: un-observability of state
  - Higher fidelity sims often require **more** state initialization
  - Do we guess? Do we say something smart? Do we measure?
  - String theory is like the extreme limit of this issue :)

Accuracy/Usefulness vs. Fidelity/Principledness

Much of the Fusion Community Seems to Implicitly Assume This



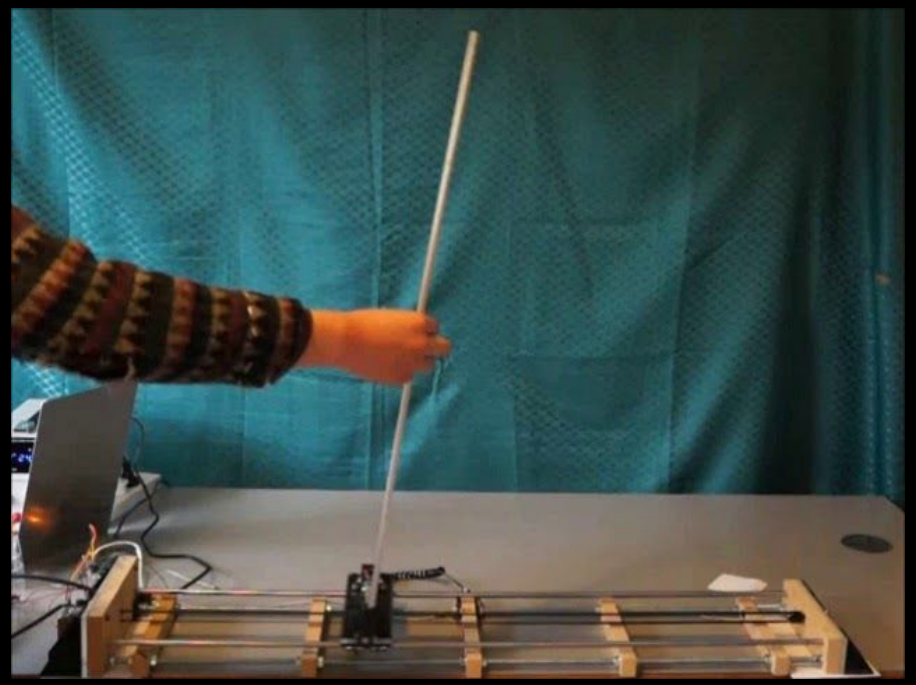
(My Anecdotal Observation)
Real-World Engineering
Suggests this is More Accurate



Learning State-Space Models from Data

Sim2Real

- Sim never equals real
- How do we reduce the gap?



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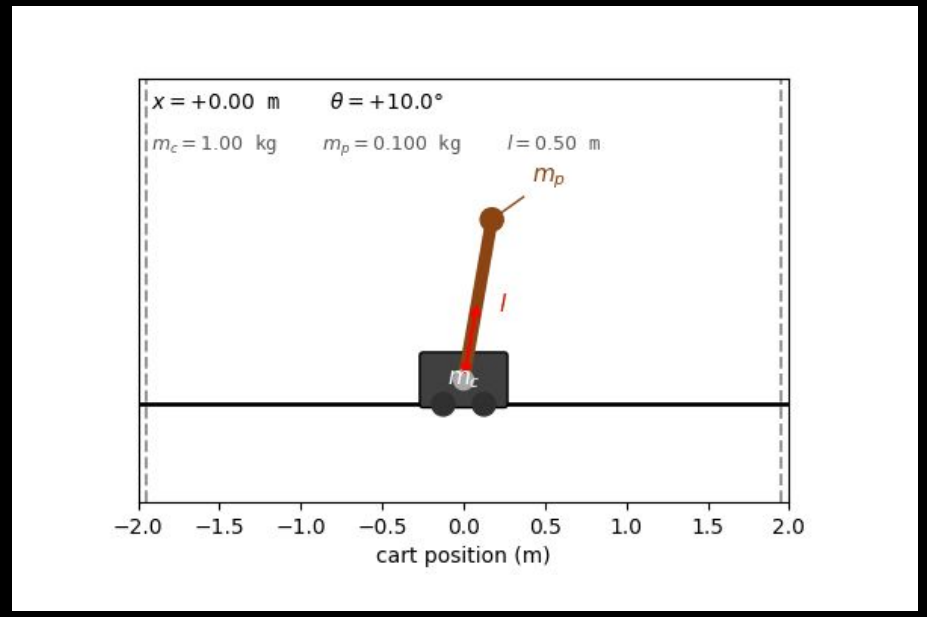


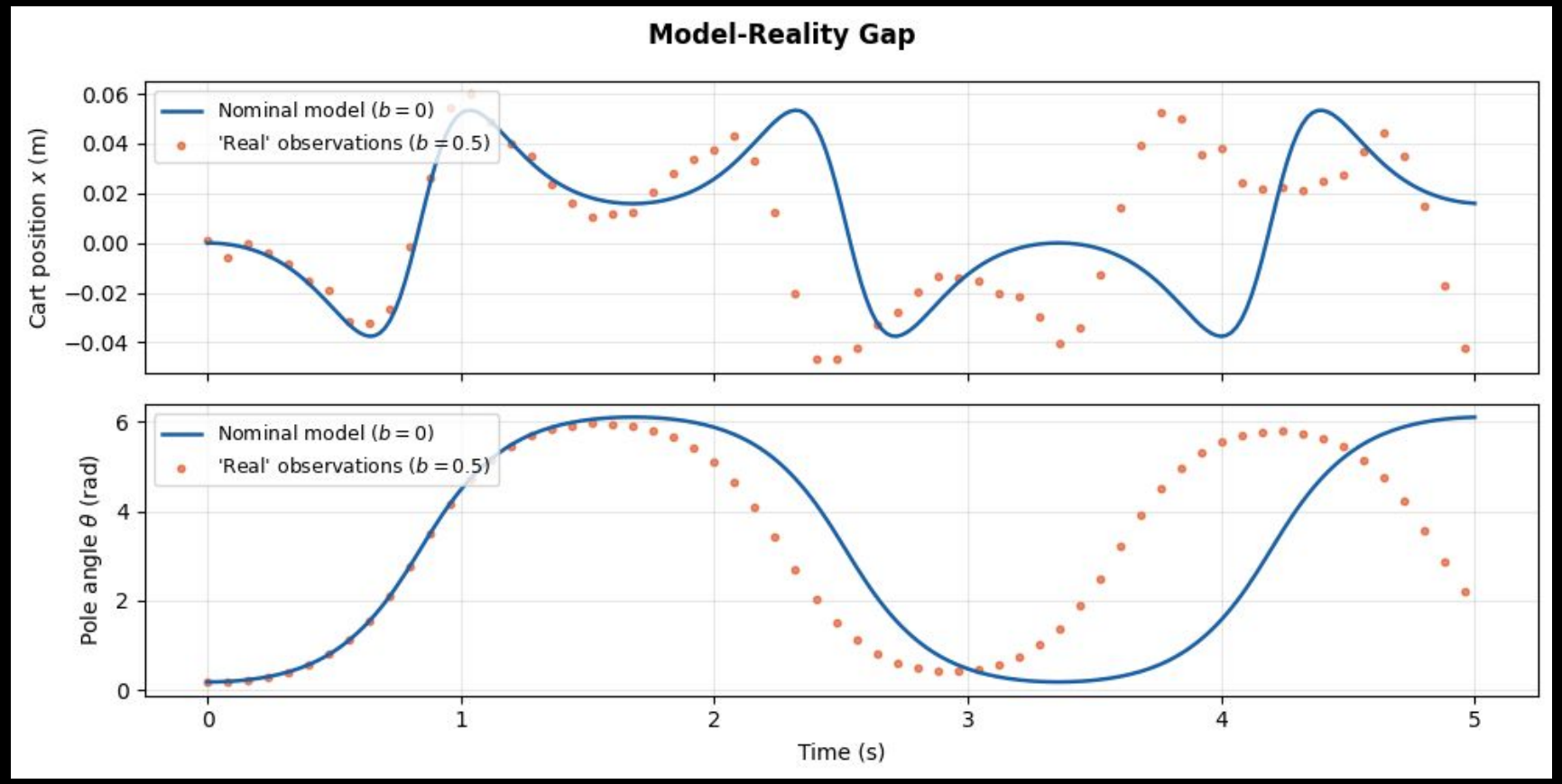
Toy Example

- We will work with a toy example
  - Assume the real cartpole has friction, according to some simple friction model

$$\ddot{x} = \frac{F \boxed{- b\dot{x}} + m_p l(\dot{\theta}^2 \sin\theta - \ddot{\theta}\cos\theta)}{M}$$

Toy Example

- Given a dataset of observations and actions, learn a correction



Learning Dynamics from Data

- Recall our state-space model, which has parameters “p”

$$\mathbf{x}_{t+1} \sim f(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$
$$\mathbf{o}_t \sim O(\mathbf{x}_t, \mathbf{a}_t; \mathbf{p})$$

- Procedure:
  - Use observations + actions to initialize states “x0”
  - Use actions to simulate a time-series of observations “o”
  - Compute error between simulated obs and real obs
  - Use auto-diff to compute d(error)/dp
  - Use your favorite optimization method to make error go down

Classical System Identification

- Learning dynamics from data has been around for many decades
- Classical approach “system identification” focuses on learning parameters for structured models
  - For our toy problem, this means estimating “b”

$$\ddot{x} = \frac{F - b\dot{x} + m_p l(\dot{\theta}^2 \sin\theta - \ddot{\theta}\cos\theta)}{M}$$

✏️Exercise - Training the Classical Model + Fiddling with Horizon Length

💡 Pause and Ponder

- System identification works great here because we **knew** the form of the missing physics and we had a parameter slot for it
  - What additional issues might the real system have?
  - What options do we have?
  - And what if friction/flex/drag change?
- **Comment:** you can view language modelling as predicting the time-series evolution of words.
  - Overwhelming empirical evidence suggests highly structured models are a bad idea for this.

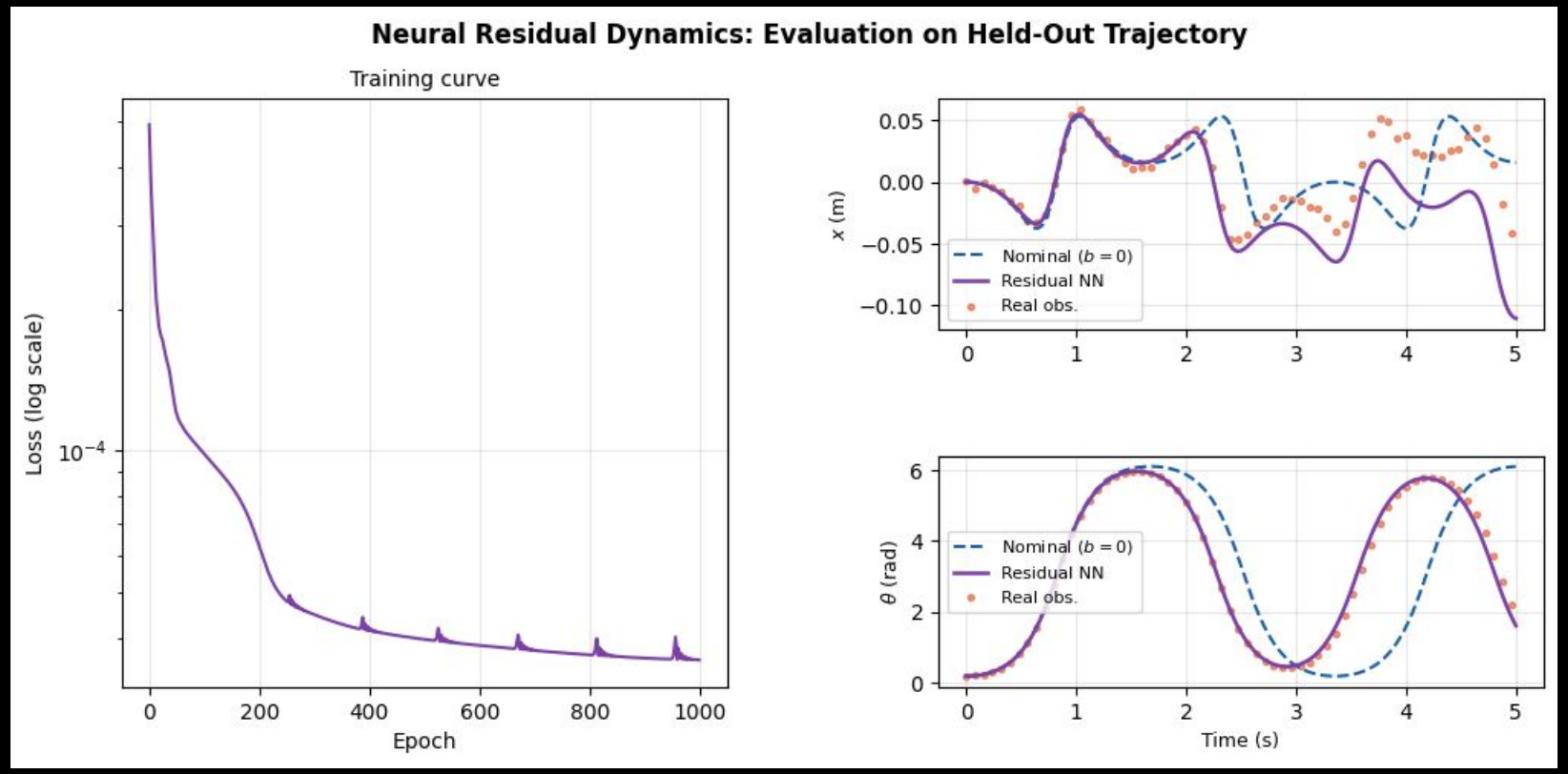
F=ma + AI, but unironically

- Another option combines physics with a network
- People often call approaches like this “SciML” (a very imprecise term)

$$\mathbf{x}_{t+1} = \underbrace{f_{\text{physics}}(\mathbf{x}_t, \mathbf{a}_t;\ \mathbf{p}_{\text{nominal}})}_{\text{known physics}} + \underbrace{f_\theta(\mathbf{x}_t, \mathbf{a}_t)}_{\text{learned correction}}$$

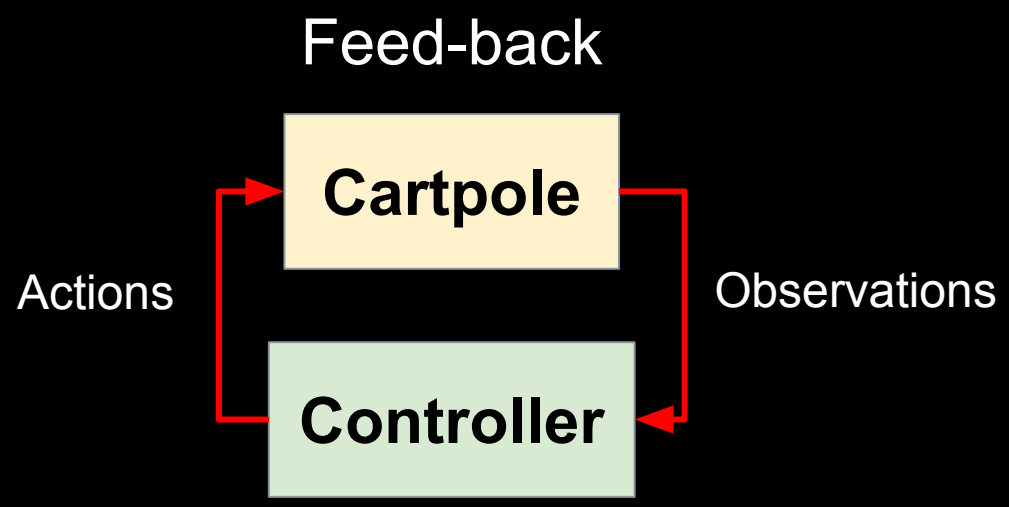
F=ma + AI, but unironically

- Noticeable improvement, but not as good as the structured model
- Why?



Real-time Estimation/Learning

- Remember that at the end of the day, we are just trying to design a control system to make the cartpole do a thing
- What if we estimated the friction in real-time?



Example: Autonomous Formula Racing

How does one even begin to tackle this with physics sim?

Tire-road interaction is complex multi-scale physics

Race-course is non-homogenous with properties changing depending on e.g. temp

Learn the Track in Real-time, Get Faster

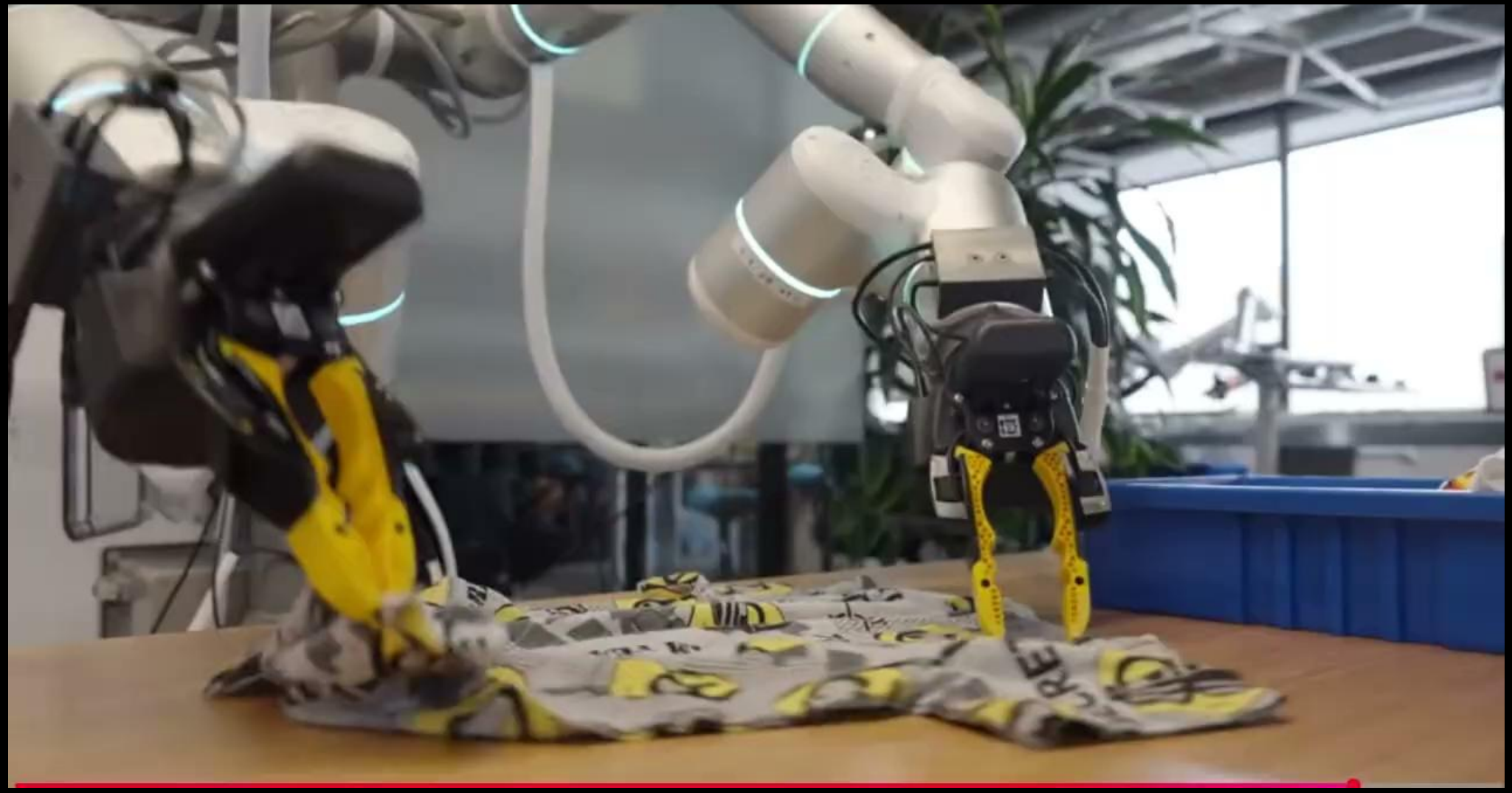
Kabzan, Juraj, et al. "Learning-based model predictive control for autonomous racing." IEEE Robotics and Automation Letters 4.4 (2019): 3363-3370.

💡 Pause and Ponder

- I just spent some time presenting a case study showing higher fidelity physics sim (i.e. tire + road modelling) is not needed / the wrong way to go about things
- Why might tire + road friction sim still be needed for this?

Learning/Bypassing State Representations

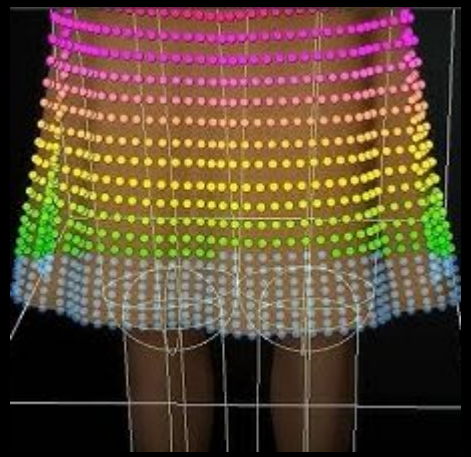
How to Simulate **this** Shirt?



How to Simulate **this** Shirt?

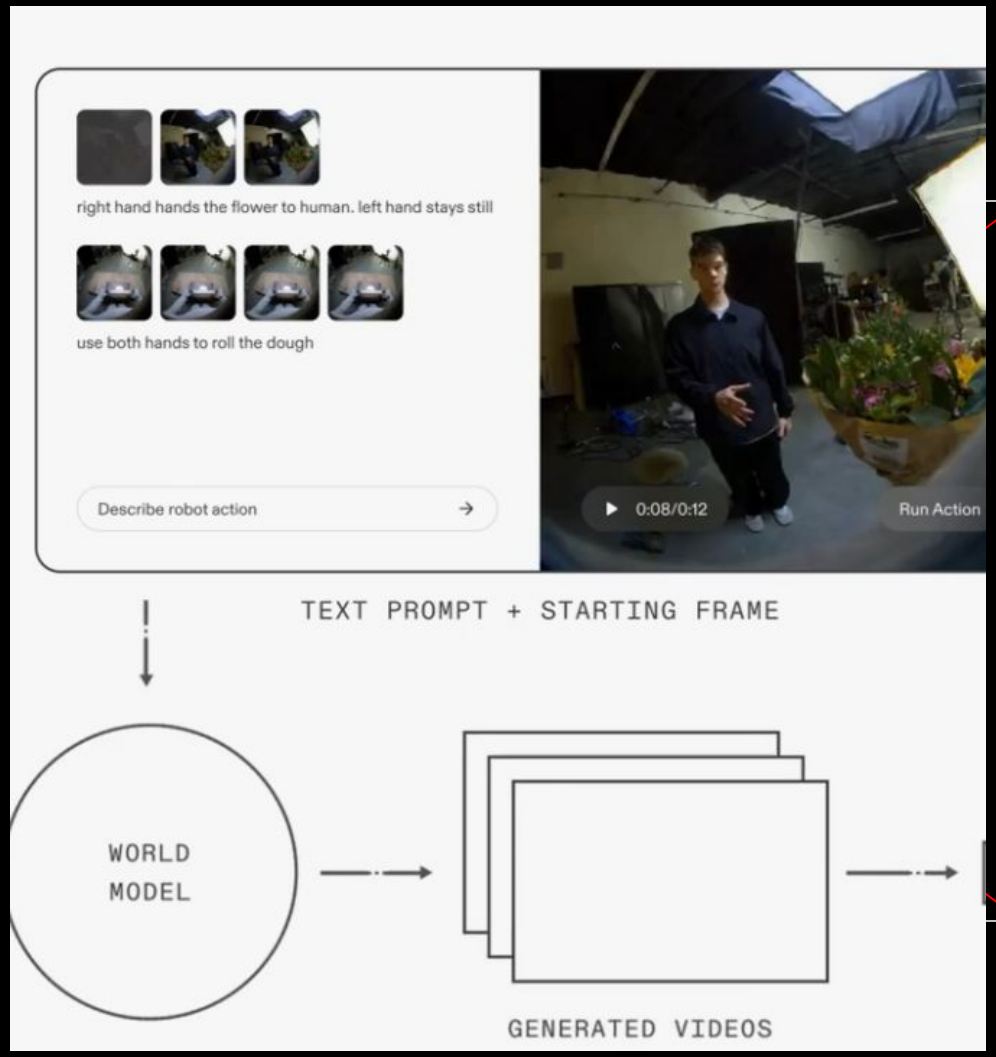
The physics approach would probably involve building some detailed mesh

The shirt is all crinkly, how does one estimate the position of all the folds? The material?



Radical Idea: Difficult State Estimation is a Limitation of Physics

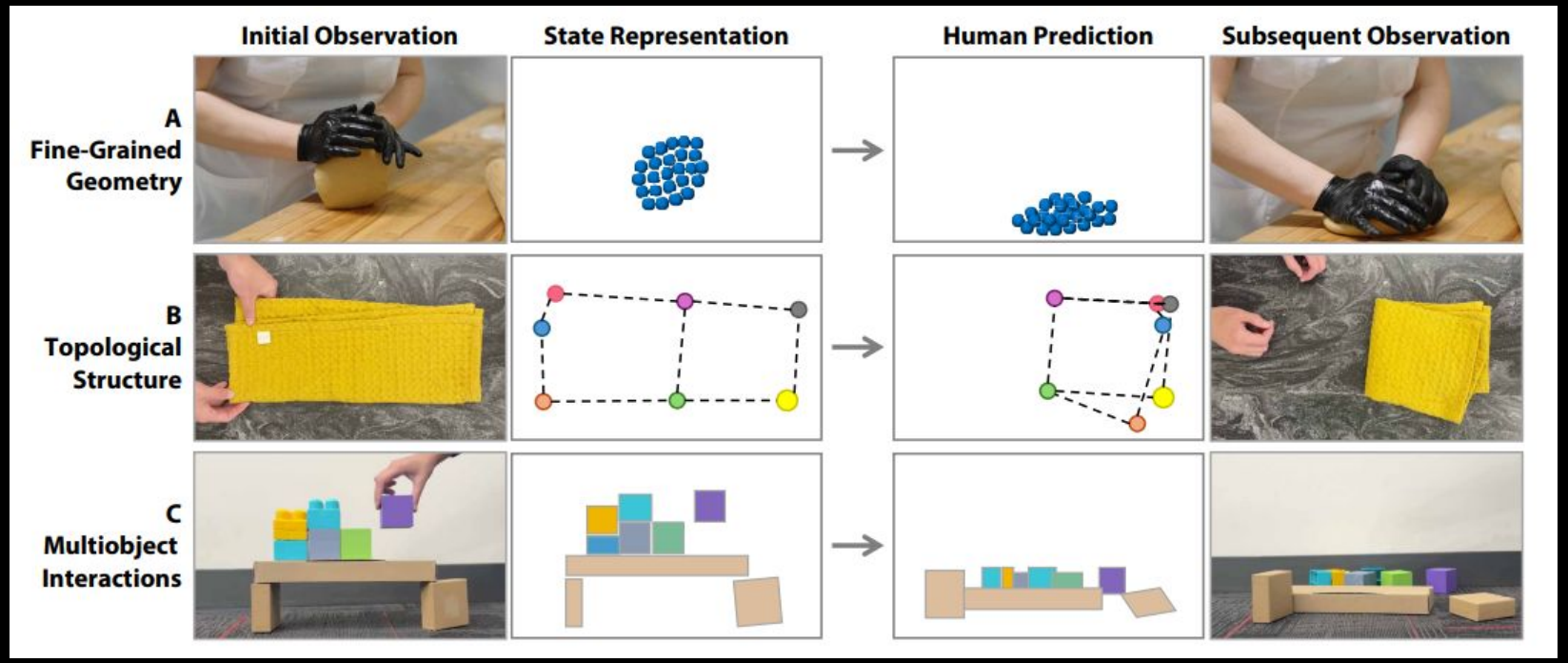
- The problem goes away if you give up on physics
- What are our observations?
  - Camera pictures + motor measurements
- What are our actions?
  - Motor commands
- Given an initial picture and motor commands, predict future pictures
  - (not quite what is happening w/ 1X world model, but see e.g. Genie 3)



1X World Model | From Video to Action: A New Way Robots Learn

Learned State Representations

- What is state estimation?
  - We have a huge vector of numbers (e.g. a picture), and we want to derive a small vector of numbers that tell us something (e.g. mass, position)
- What if we learn the state-representation?
  - Not “interpretable”, but were concepts like “charge” interpretable to you in middle school?



Ai, Bo, et al. "A review of learning-based dynamics models for robotic manipulation."
Science Robotics 10.106 (2025): eadt1497.

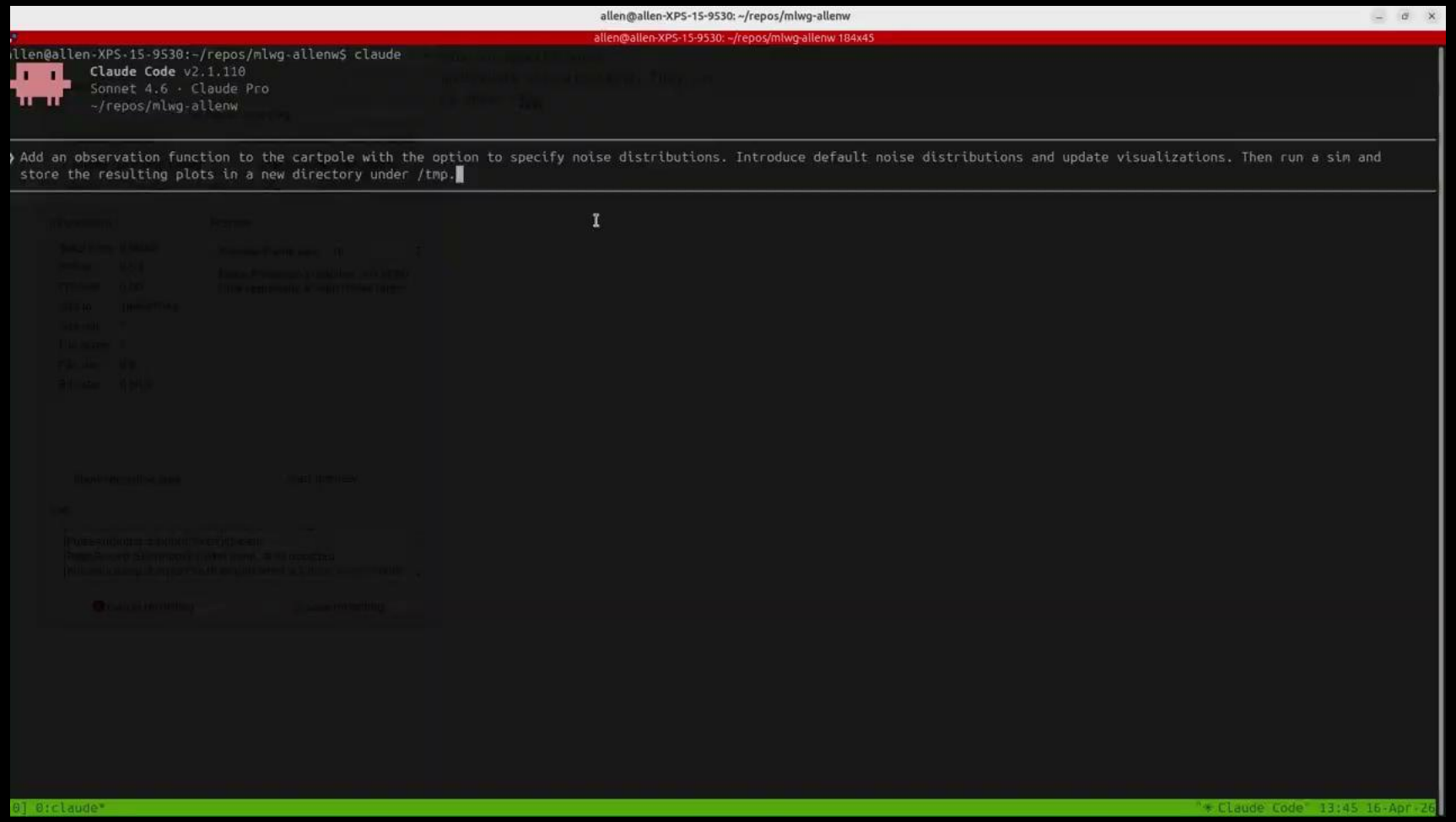
Learning/Bypassing State Representations

- In areas of robotics where simulating the system is really hard (e.g. clothes), we're starting to see models that learn state representations (or completely bypasses state) start to show signs of life
- We often struggle to estimate physics state in fusion
  - The idea of learning state representations should be taken seriously
- Fusion has the challenges of limited data, the exact same methods likely won't work
- Some more moderate version of learned state will work
  - (IMO, concepts like H-mode/I-mode/L-mode are a human-constructed state representation from empirical data)

Notes on LLM Agents

LLM Agents

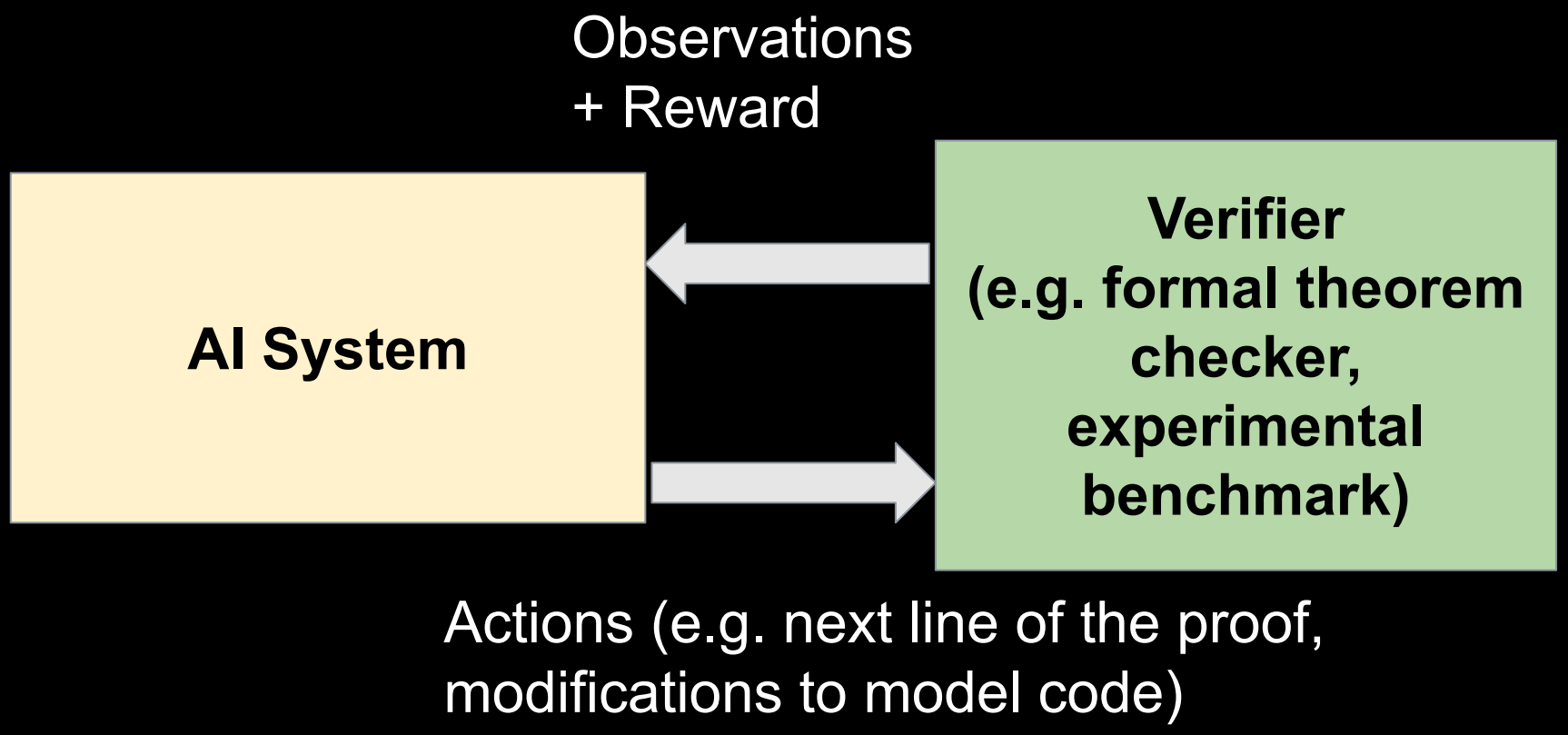
- Agents: things that interact with your computer and iterate



LLM Agents

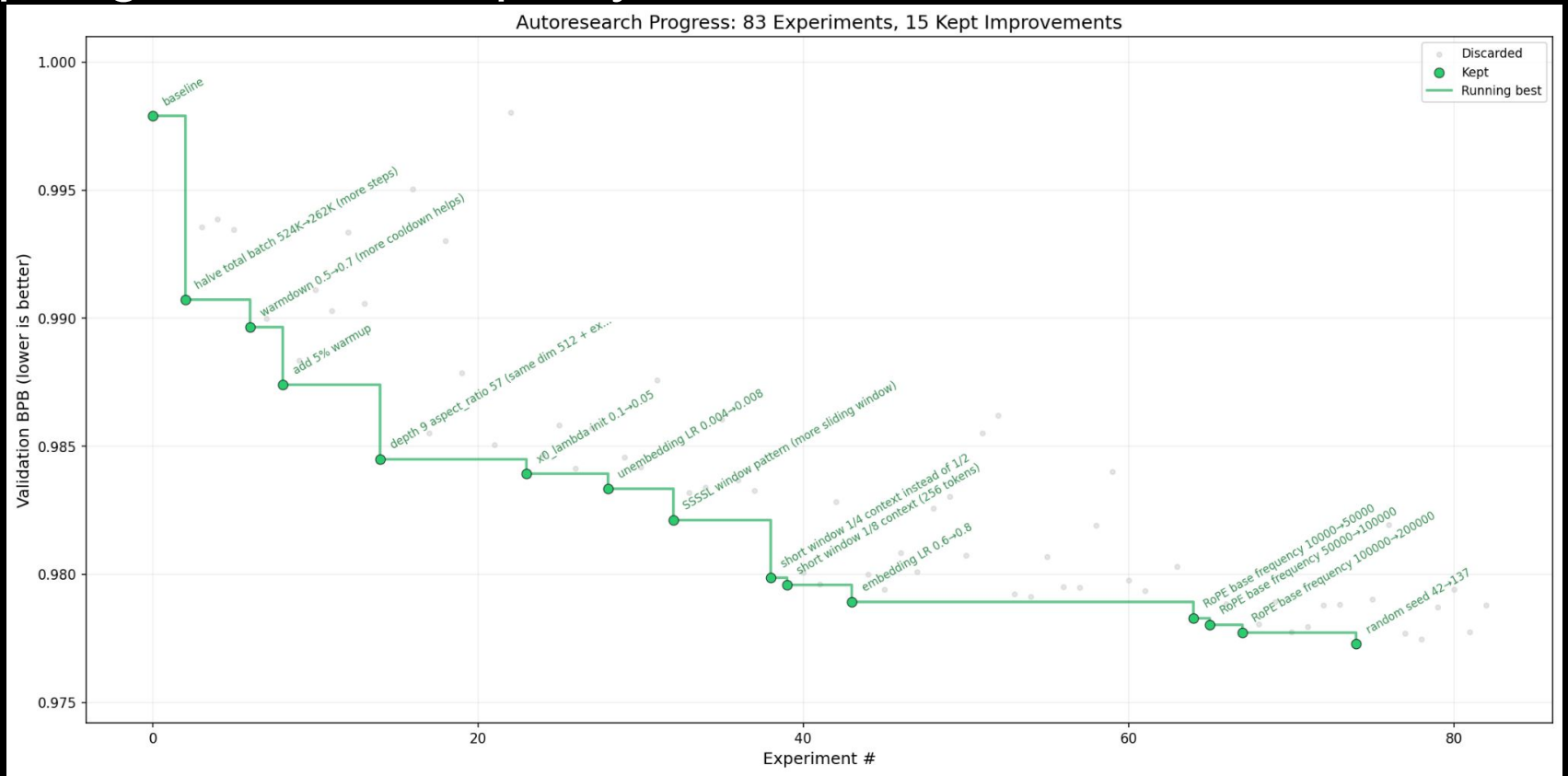
- Lots of work and oversight needed
  - Overall speed-up feels more like 2-3x than 10x or 100x
  - Good software principles act as guardrails to make agents more reliable + useful (e.g. git is **absolutely crucial**)
- Change has been fast: LLM agents were not very useful last Fall, and are now decisively very useful
- It's not just "software engineering", everything that can be done with a computer can be tackled with a coding agent
  - They're getting decent at data analysis tasks now
- The scientific world needs to figure out how to handle the onslaught of AI generated content
  - Verification + validation has never been more important
- It's not all "slop", they're actually good but w/ low reliability

Verifiable Problems are Increasingly Solvable



LLM Agents: Autoresearch

- Idea: have the agent tweak the LLM model code to improve model performance metrics
  - Andrej Karpathy showed this is starting to work a bit: https://github.com/karpathy/autoresearch



First Proof Challenge

- Authors used standard LLMs out of the box
- Various AI research groups applied custom techniques (“scaffolds”)

arXiv:2602.05192v1 [cs.AI] 5 Feb 2026

First Proof

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Andrew J. Blumberg
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Martin Hairer
EPFL and Imperial

Joe Kileel
University of Texas at Austin

Tamara G. Kolda
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Lauren Williams§
Harvard University

February 6, 2026

Abstract

To assess the ability of current AI systems to correctly answer research-level mathematics questions, we share a set of ten math questions which have arisen naturally in the research process of the authors. The questions had not been shared publicly until now; the answers are known to the authors of the questions but will remain encrypted for a short time.

AI Advances this Year + Implications for Science

- Author prompted out-of-the box LLMs got 2/10 right
  - Caveat: they chose the best LLM answer out of a sea of BS
- AI labs claimed ~5/10 right (dust needs to be settled, some caveats such as human expert guidance)
- Implications
  - At a minimum, some amount of weak generalization is occurring, with capabilities surpassed only by top human experts
  - Most attempts result in smart sounding BS generated, **verification + testing are going to be key going forward**

A.9 Question 9: Joe Kileel

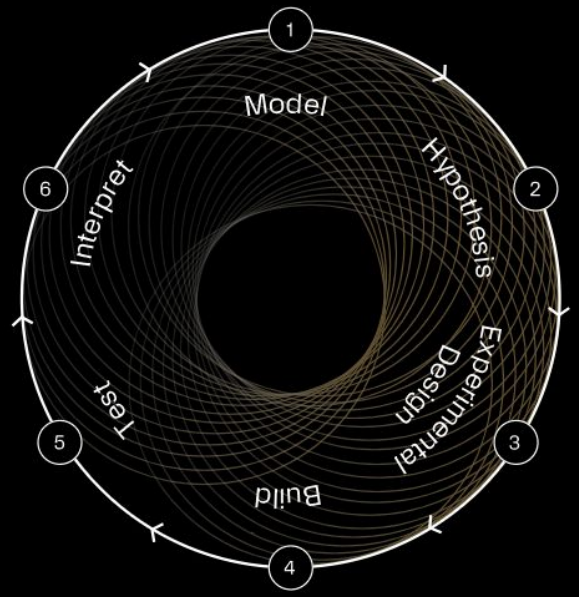
The best LLM answer found during testing was NoInternet-040226. This is an essentially correct answer. It constructs the same algebraic relations as in my own answer, namely the various 5×5 minors of the four $3n \times 27n^3$ flattenings of the block tensor assembling together the $Q^{(\alpha\beta\gamma\delta)}$. The proof by the LLM that the algebraic relations satisfy the desired properties differs from my own argument. The LLM considers a torus action on an appropriate Grassmannian, argues the stabilizer of a generic point is 1-dimensional, and uses this to show separability of λ in a somewhat

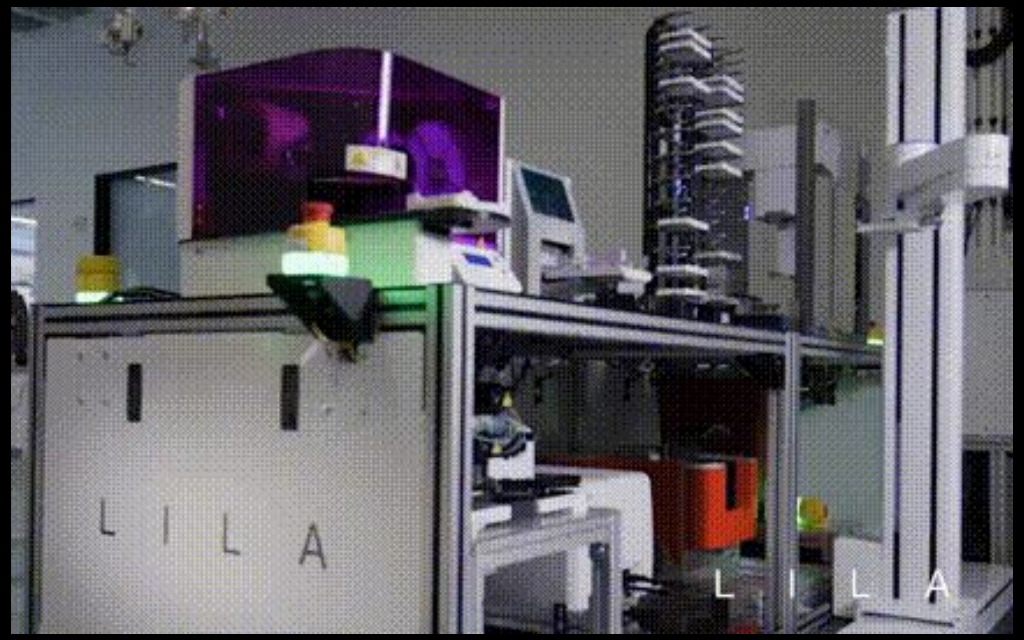
A.10 Question 10: Tammy Kolda

The best LLM solution was correct and better than the solution I provided in that it lowered the computational complexity. Most importantly, it had an insight that was obvious in hindsight but that I had not seen yet myself. Since LLMs are well known to surface existing solutions, I tried search on "subsampled kronecker product matvec" and found that the main idea in the solution exists in `https://arxiv.org/pdf/1601.01507`. (I am not sure if this is the only source of the solution, but it is at least one such solution.)

Advertisement

- What happens if we build self-driving labs?
  - AI system designs experiments, robots run them, data goes back to AI system
  - The robotic lab also provides a verification platform in general
- I'll be looking into this for materials/life science at my next job at Lila Sciences





Some Quotes

“It doesn't matter how beautiful your theory is, it doesn't matter how smart you are. If it doesn't agree with experiment, it's wrong” - Richard Feynman

“All models are wrong; some are useful” - George Box

“Scientific theories are not discoveries of the laws of nature but rather inventions of the human mind” - Athanasios Papoulis

“Scientific inquiry includes creating a testable hypothesis through inductive reasoning, testing it through experiments and statistical analysis, and adjusting or discarding the hypothesis based on the results.” - Wikipedia on the Scientific Method

AI Advances this Year + Implications for Science